



Problem 3 – Calculation of stresses using equivalent load method

The cross-section of a post-tensioned concrete beam spanning 15 m, simply supported, is shown in Figure 3.1. Beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 300 mm. The tendon comprises of 8 number 12.7 mm strands. Tendon is stressed to 80% of breaking load. Properties of the materials are:

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

$$\text{specific weight of concrete} = 25 \text{ kN} / \text{m}^3$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$f_{py} = 1640 \text{ MPa}$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

Using the equivalent load method determine:

1. Stresses in the beam section at midspan due to prestress acting alone
2. Stresses at the midspan section due to prestress, self-weight and uniform live load of 18 kN/m.
3. Is the beam cracked under this loading?

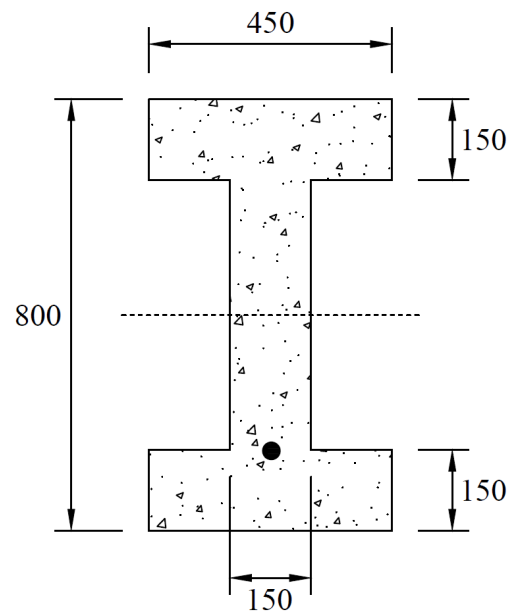


Figure 1.1 - Post-tensioned beam section

Solution

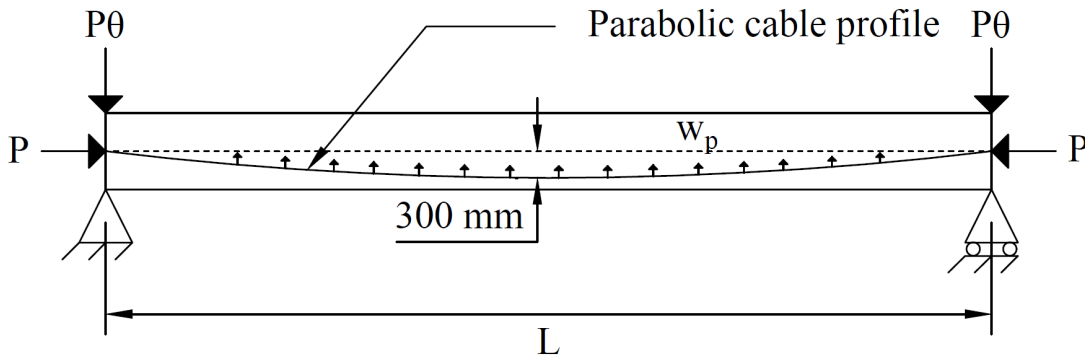


Figure 3.2 - Cable profile

$$P = 8 \times 184 \times 80\% = 1177.6 \text{ kN}$$

Equation of cable profile:

$$y = 4h \left(\frac{x}{L} - \left(\frac{x}{L} \right)^2 \right)$$

$$\frac{dy}{dx} = \theta = \frac{4h}{L} \left(1 - \frac{2x}{L} \right)$$

$$x = 0, \quad \theta = \frac{4h}{L} = \frac{4 \times 0.30}{15} = 0.08 \text{ rad}$$

Downward force at each anchorage:

$$P\theta = 0.08 \times 1177.6 = 94.2 \text{ kN}$$

↑udl due to prestress:

$$w_p = \frac{8Ph}{L^2} = \frac{8 \times 1177.6 \times 0.3}{15^2} = 12.56 \text{ kN / m}$$

$$M_{\max} = -94.2 \times \frac{15}{2} + 12.56 \times \frac{15}{2} \times \frac{15}{4} = -353.3 \text{ kNm}$$

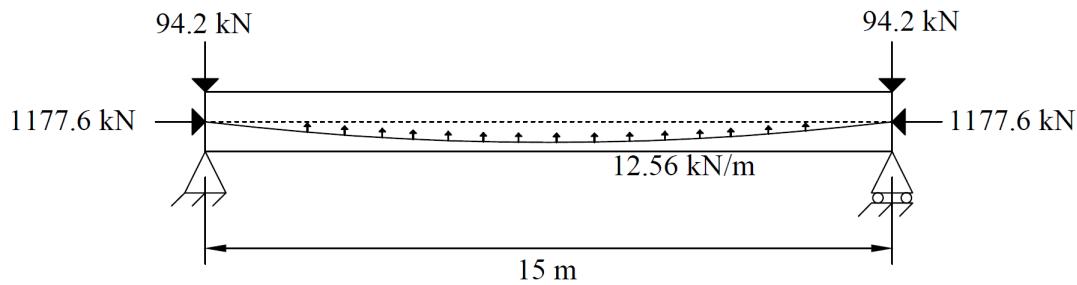


Figure 3.3 – Beam loading

You can obtain the same value using:

$$M_{\max} = \frac{w_p L^2}{8} = \frac{12.56 \times 15^2}{8} = 353.3 \text{ kNm}$$

1. Extreme fibre stresses due to prestress alone

Top fibre stress:

$$\sigma_t = \frac{P}{A_g} - \frac{My}{I_g}$$

$$\sigma_t = \frac{1177.6 \times 10^3}{210,000} - \frac{353.3 \times 10^6 \times 400}{1.61 \times 10^{10}} = 5.61 - 8.78 = -3.2 \text{ MPa}$$

Bottom fibre stress:

$$\sigma_b = \frac{P}{A_g} + \frac{My}{I_g}$$

$$\sigma_b = 5.6 + 8.8 = 14.4 \text{ MPa}$$

Note these values are consistent to those in Question 2.

2. Stresses at mid-span due to prestress, self-weight and uniform live load = 18 kN/m:

Net load can be calculated as:

Self-weight = 5.25 kN / m ↓

Live load = 18 kN / m ↓

Prestress equivalent load = 12.56 kN / m ↑

Net load = 5.25 + 18 - 12.56 = 10.69 kN / m ↓



$$M_{\max} = \frac{10.7 \times 15^2}{8} = 301 \text{ kNm}$$

$$\text{stress due to applied BM} = \frac{My}{I} = \frac{301 \times 10^6 \times 400}{1.61 \times 10^{10}} = 7.47 \text{ MPa}$$

$$\sigma_t \text{ due to applied BM} = +7.47 \text{ MPa}$$

$$\sigma_b \text{ due to applied BM} = -7.47 \text{ MPa}$$

Stress due to prestress and BM:

$$\sigma_t = 5.6 + 7.47 = 13.1 \text{ MPa}$$

$$\sigma_b = 5.6 - 7.47 = -1.87 \text{ MPa}$$

3. *Is the beam cracked?*

$$\text{Since, } -0.6\sqrt{f'_c} = -0.6\sqrt{32} = -3.39 \text{ MPa}$$

$$-3.3 \text{ MPa} < -3.39 \text{ MPa}$$

The beam is uncracked.



Problem 4 – Stresses in prestressed one-way slab using equivalent load method

A 220 mm thick slab spans 10 m. It will be post-tensioned with regularly spaced tendons with parabolic profiles and zero eccentricity at end anchorages. To limit the deflection of the slab, the designer has decided to balance the self-weight of the slab completely by the applied prestress.

Material properties are:

$$f'_c = 50 \text{ MPa}$$

$$f'_{cp} = 40 \text{ MPa}$$

Strands of nominal diameter 12.7 mm

$$A_p = 800 \text{ mm}^2$$

$$f_p = 1870 \text{ MPa}$$

$$f_{py} = 1640 \text{ MPa}$$

Tendon stressed to 70% of breaking load.

Assuming that the minimum cover required for the tendon is 25 mm, and that immediate losses are 5% of the jacking force and time-dependent losses are 17% of the force at transfer;

1. Find the prestress force required to balance the self-weight
2. Check whether the slab can sustain the applied prestress at transfer.

Solution

1. Prestress force required to balance the self-weight.

Load to be balanced = self-weight

Consider 1 m width of the slab and treat it as a beam.

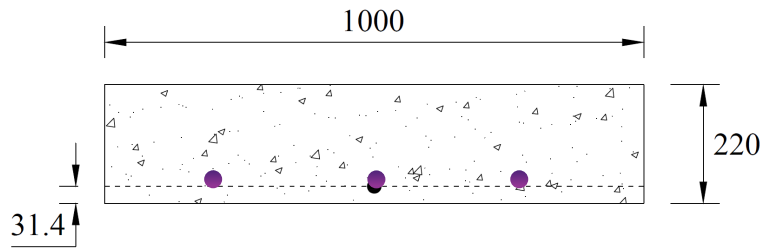


Figure 4.1 - One-way slab section

Assuming parabolic cable profile:

$$e = \frac{220}{2} - 31.4 = 78.6 \text{ mm}$$

$$\text{self-weight} = 0.22 \times 1 \times 25 = 5.5 \text{ kN / m}$$

Upward UDL required from prestress = 5.5 kN / m

$$w_p = \frac{8Pe}{L^2} \quad \text{where } h = e$$

$$P_e = \frac{w_p L^2}{8e} = \frac{5.5 \times 10^2}{8 \times 0.0786} = 875 \text{ kN}$$

This is the prestress force required during service, after all prestress loss has occurred.

To allow for 17% time-depended losses we obtain the initial prestress force:

$$P_i = \frac{875}{0.83} = 1054 \text{ kN}$$

To allow for 5% immediate losses at jacking, the jacking force is:

$$P_j = \frac{1054}{0.95} = 1110 \text{ kN / metre width}$$

If a four strand cable containing 12.7 mm ϕ super grade strands stressed to 70% is used, the force taken by one cable:

$$4 \times 184 \text{ kN} \times 70\% = 515 \text{ kN}$$

$$\text{Spacing of cables required in the slab} = \frac{1000}{1110} \times 515 = 463 \text{ mm ie use 450 mm}$$

Reduce to three strand cable

$$3 \times 184 \times 70\% = 386 \text{ kN}$$

New spacing of cables required in the slab

$$\text{Spacing of cables required in the slab} = \frac{1000}{1110} \times 386 = 348 \text{ mm ie use 300 mm}$$

Therefore adopt $3 \times 12.75 \text{ mm}$ strands at 300 mm spacing.

2. Check prestress at transfer

At transfer the slab will support prestress and self-weight.

$$\frac{\text{prestress}}{\text{metre width}} = \frac{386}{300} \times 1000 = 1287 \text{ kN}$$

At transfer, 5% immediate losses have occurred:

$$P_i = 1287 \times 0.95 = 1223 \text{ kN}$$

$$w_p = \frac{8Pe}{L^2} = \frac{8 \times 1223 \times 0.0786}{8^2} = 7.73 \text{ kN/m} \uparrow$$

$$\text{Net load} = w_p - w_{sw} = 7.73 - 5.5 = 2.3 \text{ kN/m} \uparrow$$

$$I_g = \frac{1}{12} \times 1000 \times 220^3 = 8.87 \times 10^8 \text{ mm}^4$$

$$y = \frac{220}{2} = 110 \text{ mm}$$

$$M_{net} = \left(\frac{2.3 \times 8^2}{8} \right) = 28.7 \text{ kNm}$$

$$\sigma_t = \frac{P}{A_g} - \frac{My}{I}$$

$$\sigma_t = \frac{1287 \times 10^3}{1000 \times 220} - \frac{29 \times 10^6 \times 110}{8.87 \times 10^8}$$

$$\sigma_t = 5.87 - 3.56 = 2.29 \text{ MPa (compression)}$$

$$\sigma_b = 5.87 + 3.56 = 9.43 \text{ MPa (compression)}$$

Since top and bottom fibres are under compression, the applied prestress is OK.